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A Phenomenological Approach to Micellisation Kinetics

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A general phenomenological treatment of micellisation kinetics applicable to solutions containing mixed micelles is developed. Large and small perturbations from equilibrium are considered with the main emphasis on the latter where linear methods apply. It is argued that the micelle–monomer exchange kinetics of each micellar species can be described in terms of a single driving force, namely the chemical potential difference between micellar and monomeric forms. The approximations inherent in this viewpoint are discussed for single-component micelles. The particular case of alcohol solubilisation is considered and an expression is obtained for the relaxation time which enables equilibrium thermodynamic data to be used in the interpretation of the kinetic results. The driving force for the slow process due to changes in micelle number is shown to be the subdivision potential both for the stepwise association and micellar association mechanisms. Also, for single-component micelles the phenomenological coefficient is shown to be the equilibrium rate of micelle formation and breakdown divided by RT. Expressions are derived which allow for coupling between the fast and slow processes. The treatment leads to the same results as previous work based mainly on mass-action kinetics when the particular features of the latter are fed into it. However, it is more general and in some respects more convenient to use.

In a recent paper[1] the stepwise association model of micellisation developed by Aniansson and coworkers[2–4] was modified by allowing for the changes in rate coefficients with solution composition that occur in solutions of ionic surfactants. This was done by postulating that the specific rates of the reactions concerned depend only on a small number of quantities which can be varied independently at equilibrium. The resultant treatment was also applied to mixed micelles.

In the same paper a brief outline was given of an alternative approach based on the thermodynamics of irreversible processes.[5] This approach is considered more fully in the present paper. It gives results which are identical to those obtained previously and has two major advantages. It is more general and requires less tedious algebraic manipulation. A number of new problems are tackled formally. Among these are (1) a more complete description of the fast process for mixed micelles than that given previously, (2) a more general description of the slow process, (3) coupling between the two processes and (4) an extension to deviations from equilibrium beyond the linear regime.

Whenever possible the thermodynamic aspects of the treatment are handled generally. As a consequence the treatment appears somewhat formal. However, this drawback is offset by the advantage that any particular thermodynamic treatment can be used.

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Basic Kinetic and Thermodynamic Equations

We consider a system consisting of a solvent, component *o*, a series of micellar species, *r*, *s*, *t*, *etc.*, and a series of non-micellar species *i*, *j*, *k*, *etc.*, which include among them the non-micellar forms of the micellar constituents. According to the stepwise association model the only reactions of any significance in the formation and breakdown of micelles are the uptake and release of monomeric material. A typical reaction may be written as

$$r+i \rightleftharpoons s$$

where in this instance *s* is the micellar species formed by the addition of an *i* molecule or ion to a micelle of type *r*. According to linear phenomenological theory the net rate of the above reaction, J_{ri}, is given by

$$J_{ri} = l_{ri}(\theta_r+\theta_i-\theta_s) \tag{1}$$

where the subscript *ri* denotes the particular reaction concerned, l_{ri} is the appropriate phenomenological coefficient and the corresponding driving force $(\theta_r+\theta_i-\theta_s)$ is the affinity of the above reaction. For uncharged species the θ_i are chemical potentials as usually understood. For charged species θ_i is the chemical potential of the species concerned plus the chemical potential of enough counterions to neutralise its charge. Further details concerning this notation and usage are given in ref. (6)–(8), and it is sufficient for our present purposes to note that in systems containing ions the θ_i play the same role as the chemical potentials μ_i in systems where all species are uncharged.

Since the above reaction is an elementary process we may write[9, 10]

$$l_{ri} = J^*_{ri}/\boldsymbol{k}T \tag{2}$$

where J^*_{ri} is the equilibrium forward or backward rate of the above reaction. Eqn (2) is equally valid in ideal and non-ideal systems and does not rely on the reactions conforming to transition-state theory.[11]

Eqn (1) and its analogues for all other micellar and monomeric species are the basic kinetic equations on which the analysis of both fast and slow processes for the stepwise association model is based.

To describe the thermodynamics of states undergoing reaction we use a generalised form of the procedures outlined in the appendix of ref. (1). We begin with the fundamental equation of the above system, which at constant T and p may be written as

$$\mathrm{d}g = \sum_i \theta_i \,\mathrm{d}m_i + \sum_r \theta_r \,dC_r \tag{3}$$

where g denotes the Gibbs free energy per mole of solvent, m_i is the concentration of non-micellar i and C_r is the concentration of micellar species r. The units of m_i and C_r are moles per mole of solvent and differ from molalities only by a constant multiplying factor.

Let N_i^r denote the number of i molecules or ions present in a micelle of type r. Let $C_i^{\mathrm{m}} = \sum_r N_i^r C_r$ denote the concentration of micellar i and let $C_{\mathrm{m}} = \sum_r C_r$ denote the concentration of micelles. When g has its minimum value consistent with given values of the m_i, C_i^{m} and C_{m} it is readily shown by the method of undetermined multipliers that the θ_r are given by

$$\theta_r = \sum_i N_i^r \,\theta_i^{\mathrm{m}} + \varepsilon \tag{4}$$

where the multipliers θ_i^{m} and ε are the same for all micellar species.[1, 12] Eqn (4) enables eqn (3) to be rewritten as

$$\mathrm{d}g = \sum_i \theta_i \,\mathrm{d}m_i + \sum_i \theta_i^{\mathrm{m}} \,\mathrm{d}C_i^{\mathrm{m}} + \varepsilon \,\mathrm{d}C_{\mathrm{m}}. \tag{5}$$

It is evident from this equation that we may regard θ_i^m as the chemical potential of micellar i and also that ε is the subdivision potential defined by Hill.[13]

When eqn (4) and (5) apply the micelles are in equilibrium among themselves but not necessarily with monomer. Also C_m does not in general have the value which for given m_i and C_i^m minimises g. We use the term 'partial equilibrium' to describe such states. When $\theta_i^m = \theta_i$ for all i the system is in chemical equilibrium and when $\varepsilon = 0$ we have subdivision equilibrium. When both conditions are satisfied the system is in complete equilibrium.

When a micellar system in complete equilibrium is perturbed as in a T-jump or p-jump experiment both chemical and subdivision equilibria are disturbed. According to the stepwise association model the return to equilibrium takes place in essentially two stages. The faster stage is the re-equilibration of monomeric material with micelles and takes place at approximately constant C_m. The slower process is the establishment of the new equilibrium value of C_m which takes place under the condition that chemical equilibrium is maintained throughout. Thus whereas eqn (1) suggests that there should be a relaxation time corresponding to each independent elementary reaction, in practice one observes at most a single relaxation process corresponding to the equilibration of each micellar constituent and a slower relaxation process corresponding to the equilibration of C_m. This suggests that for a given micellar constituent those relaxation processes which contribute significantly to the overall amplitude are kinetically alike. We proceed as follows.

Treatment of the Fast Process

Derivation of General Equations for Relaxation Times

For a given micellar constituent i we sum eqn (1) over all micellar species to obtain

$$\frac{dm_i}{dt} = \sum_r l_{ri}(\theta_s - \theta_r - \theta_i). \tag{6}$$

We now rewrite this expression as

$$\frac{dm_i}{dt} = \sum_r l_{ri}[(\theta'_s - \theta^*_s) - (\theta'_r - \theta^*_r) - (\theta'_i - \theta^*_i)] + \sum_r l_{ri}[(\theta_s - \theta'_s) - (\theta_r - \theta'_r) - (\theta_i - \theta'_i)] \tag{7}$$

in which superscript * denotes the state at the end of the fast process in which we have chemical but not subdivision equilibrium and superscript $'$ denotes the partial equilibrium state with the same C_i^m m_i and C_m as the perturbed state of interest. It is apparent from eqn (4) that $(\theta'_s - \theta'_r) = \theta_i^m$ is the same for all micellar species and that $(\theta^*_s - \theta^*_r) = \theta^*_i$. Consequently when the second term on the right-hand side of eqn (7) is negligible compared with the first this equation simplifies to give

$$\frac{dm_i}{dt} = \sum_r l_{ri}[(\theta_i^m - \theta^*_i) - (\theta'_i - \theta^*_i)] \tag{8}$$

which is of linear phenomenological form with a driving force $(\theta_i^m - \theta'_i)$ and a phenomenological coefficient given by $\sum_r l_{ri}$. We note that $\boldsymbol{RT}\sum_r l_{ri}$ is the equilibrium rate at which i monomers enter or leave the micelles. Now according to eqn (5) the differences $(\theta_i^m - \theta^*_i)$ and $(\theta'_i - \theta^*_i)$ in eqn (8) can be expressed in terms of the perturbations in the m_i and the C_i^m. Hence we have

$$(\theta_i^m - \theta^*_i) = \sum_k \left(\frac{\partial \theta_i^m}{\partial m_k}\right)_{m_j, C_k^m, C_m} \Delta m_k + \sum_k \left(\frac{\partial \theta_i^m}{\partial C_k^m}\right)_{m_k, C_j^m, C_m} \Delta C_k^m \tag{9a}$$

$$(\theta'_i - \theta^*_i) = \sum_k \left(\frac{\partial \theta'_i}{\partial m_k}\right)_{m_j, C_k^m, C_m} \Delta m_k + \sum_k \left(\frac{\partial \theta_i^m}{\partial C_k^m}\right)_{m_k, C_j^m, C_m} \Delta C_k^m \tag{9b}$$

where the prefix Δ denotes a quantity in the perturbed state of interest minus the corresponding quantity at the end of the reaction. The derivatives are taken at constant C_m because C_m is assumed not to change significantly during the fast process.

Since the perturbations occur in a closed system we have for each constituent

$$\Delta m_k + \Delta C_k^m = 0. \tag{10}$$

Consequently eqn (8) becomes

$$\frac{dm_i}{dt} = [\sum_r l_{ri}] \sum_k \left[\left(\frac{\partial \theta_i^m}{dm_k} \right)_{m_j, C_k^m, C_m} - \left(\frac{\partial \theta_i^m}{\partial C_k^m} \right)_{m_k, C_j^m, C_m} - \left(\frac{\partial \theta_i'}{\partial m_k} \right)_{m_j, C_k^m, C_m} + \left(\frac{\partial \theta_i'}{\partial C_k^m} \right)_{m_k, C_j^m, C_m} \right] \Delta m_k. \tag{11}$$

There is an analogue of eqn (11) for each micellar constituent and the resultant set of equations lead to a set of relaxation times whose reciprocals are the eigenvalues of the matrix $|a_{ik}|$, where a_{ik} is the coefficient of Δm_k in eqn (11). In general the derivatives in eqn (11) are not obtainable from equilibrium experimental data. They can, however, be estimated if we have detailed expressions for the dependence of chemical potentials on solution composition. For solutions of non-ionic surfactants and ionic surfactants plus salt it can be shown, albeit by some rather tedious manipulation, that the same expressions for the relaxation times result as were obtained previously[1] when the same expressions for the chemical potentials are used. However, the present treatment is more general because it can accommodate situations which the previous treatment cannot. An example is the case where monomeric surfactant ions behave differently from inorganic ions of the same charge type and may specifically influence the behaviour of the micelles.

The only significant approximation made above is that the second term on the right-hand side of eqn (7) is much smaller than the first. Whether or not this is so will depend to some extent on the perturbation concerned. For single-component non-interacting micelles Almgren *et al.*[4] have derived the conditions whereby this term is zero for arbitrary perturbations. A more general discussion of the issues involved for single-component micelles is outlined in the appendix.

Application to Alcohol Solubilisation

As a particular application of the above treatment we consider the kinetics of alcohol exchange between micellar and aqueous environments in solutions of a longish chain-length ionic surfactant (C_{16} or thereabouts) and a shortish chain-length alcohol such as pentanol. For solutions of this kind one expects two well separated relaxation times associated with the micelle–monomer equilibria with the faster time attributable primarily to the alcohol exchange and the slower time associated with the surfactant exchange.

When dealing with the faster process it will be a good approximation to assume that the amount of micellar surfactant as well as the concentration of micelles remains constant. On the other hand one might expect the alcohol to be effectively in equilibrium with micelles during the slower process of surfactant exchange. Moreover, since the equilibrium concentration of surfactant monomer will be rather small the effect of the alcohol thereon can almost certainly be ignored. Under these circumstances the alcohol exchange kinetics are governed approximately by the expression

$$\frac{dm_i}{dt} = (\sum_r l_{ri}) \left[\left(\frac{\partial \theta_i^m}{\partial m_i} \right)_{C_i^m} - \left(\frac{\partial \theta_i^m}{\partial C_i^m} \right)_{m_i} - \left(\frac{\partial \theta_i'}{\partial m_i} \right)_{C_i^m} + \left(\frac{\partial \theta_i'}{\partial C_i^m} \right)_{m_i} \right] \Delta m_i \tag{12}$$

where i denotes the alcohol and the derivatives are taken at constant C_k^m, m_k and C_m, with k denoting the surfactant.

In general the derivatives on the right-hand side of eqn (12) cannot be evaluated from equilibrium measurements. However, if the chemical potential of monomeric alcohol is given by

$$\theta_i' = \theta_i^\circ(T, p) + \boldsymbol{RT} \ln m_i \tag{13}$$

then evidently $(\partial\theta_i'/\partial C_i^m)_{m_i}$ and $(\partial\theta_i^m/\partial m_i)_{C_i^m}$ are both zero. Under these circumstances eqn (12) simplifies to give

$$\frac{dm_i}{dt} = -[\sum_r l_{ri}]\left[\left(\frac{\partial\theta_i^m}{\partial C_i^m}\right)_{C_k^m, C_m} + \frac{\boldsymbol{RT}}{m_i}\right]\Delta m_i \tag{14}$$

which leads to a relaxation time $1/\tau_i$ given by

$$1/\tau_i = \sum_r l_{ri}\left[\left(\frac{\partial\theta_i^m}{\partial C_i^m}\right)_{C_k^m, C_m} + \frac{\boldsymbol{RT}}{m_i}\right]. \tag{15}$$

Consider now the quantity $(\partial\theta_i^m/\partial C_i^m)_{C_k^m, C_m}$. Although this quantity is not obtainable directly from equilibrium measurements what can be measured at equilibrium, for example from vapour pressures, is $(\partial\theta_i^m/\partial C_i^m)\, C_{k,\varepsilon}^m$ because at equilibrium ε = constant = 0. However, since

$$\left(\frac{\partial\theta_i^m}{\partial C_i^m}\right)_{C_k^m, C_m} = \left(\frac{\partial\theta_i^m}{\partial C_i^m}\right)_{C_k^m, \varepsilon} + \left(\frac{\partial\theta_i^m}{\partial\varepsilon}\right)_{C_i^m, C_m^k}\left(\frac{\partial\varepsilon}{\partial C_i^m}\right)_{C_k^m, C_m} \tag{16}$$

and

$$\left(\frac{\partial\varepsilon}{\partial C_i^m}\right)_{C_k^m, C_m} = -\left(\frac{\partial\varepsilon}{\partial C_m}\right)_{C_i^m, C_m^k}\left(\frac{\partial C_m}{\partial C_i^m}\right)_{C_k^m, \varepsilon} \tag{17}$$

and according to eqn (5)

$$\left(\frac{\partial\theta_i^m}{\partial\varepsilon}\right)_{C_i^m, C_m^k} = -\left(\frac{\partial C_m}{\partial C_i^m}\right)_{C_k^m, \varepsilon} \tag{18}$$

it follows that

$$\left(\frac{\partial\theta_i^m}{\partial C_i^m}\right)_{C_k^m, C_m} = \left(\frac{\partial\theta_i^m}{\partial C_i^m}\right)_{C_k^m, \varepsilon} + \left(\frac{\partial\varepsilon}{\partial C_m}\right)_{C_i^m, C_m^k}\left(\frac{\partial C_m}{\partial C_k^m}\right)^2_{C_k^m, \varepsilon}. \tag{19}$$

Where the final term is necessarily positive for the system to be thermodynamically stable. Hence

$$\frac{1}{\tau_i} = [\sum_r l_{ri}]\left[\frac{\boldsymbol{RT}}{m_i} + \left(\frac{\partial\theta_i^m}{dC_i^m}\right)_{C_k^m, \varepsilon} + \left(\frac{d\varepsilon}{\partial C_m}\right)_{C_i^m, C_m^k}\left(\frac{\partial C_m}{\partial C_i^m}\right)^2_{C_k^m, \varepsilon}\right]. \tag{20}$$

$\sum_r l_{ri}$ is in effect the equilibrium rate of alcohol exchange between micellar and aqueous environments and can be expected to depend on the amount of alcohol present in the system. Thus it makes sense to write

$$\sum_r l_{ri} = kC_i^m/\boldsymbol{RT} \tag{21}$$

where the rate coefficient k may be regarded as the average time interval spent by an alcohol molecule in a micelle.

Eqn (20) has already been used successfully to discuss the kinetics of pentanol exchange in solutions of cetyl pyridinium chloride.[14] When used in conjunction with the phenomenological treatment of relaxation data outlined previously[15] it may be possible in favourable cases to estimate both the kinetic term $\sum_r l_{ri}$ and the thermodynamic term $(\partial\varepsilon/\partial C_m)_{C_i^m, C_k^m}\,(\partial C_m/\partial C_i^m)_{C_k^m, \varepsilon}$. This latter term is closely related to the effect of the alcohol on the number of surfactant ions per micelle and vanishes when no effect of this

kind occurs. When the kinetics of surfactant exchange are much slower than those of the alcohol it will be reasonable to treat these by using the formalism applicable to single-component micelles and regarding the influence of the alcohol as a medium effect. This viewpoint has much in common with the treatment of counterion effects described in ref. (1).

The advantages of the present approach over alternative published treatments are clear. No assumptions concerning the form of the micelle population are required. Nor need it be assumed that the micelles are non-interacting. This contrasts with the treatment of Aniansson[16] as applied by Yiv *et al.*[17] and subsequently extended by Wall and Elvingson.[18] Also, no particular forms need be assumed for the law describing the partitioning of the alcohol between micellar and aqueous environments or for the kinetic equations describing the exchange between the two environments. This contrasts with previous work based on a two-state model.[19, 20] Moreover, the present treatment does not invoke any assumptions which are not either explicitly or implicitly present in the alternatives.

The above application is but a single example illustrating the power of the phenomenological approach. Similar methods are applicable to more complex micellar systems, to binding by macromolecules and possibly also to macromolecular conformation changes. However, the treatment remains approximate and it is clear that further evaluation of the approximation inherent in eqn (8) is desirable.

Linear Treatment of the Slow Process

It is generally accepted that the slow relaxation process is concerned with equilibration of C_m the concentration of micelles whilst the micelles remain effectively in equilibrium with monomer. In thermodynamic language this corresponds to the attainment of subdivision equilibrium under conditions of chemical equilibrium. According to the stepwise association model this process can only occur through the breakdown or formation of micelles by loss or gain of monomers. More recently evidence has been forwarded which suggests that micellar association and disproportionation are important in some cases.[21, 22] In the discussion below both mechanisms will be considered.

For the stepwise association model we follow closely the arguments given by Aniansson and Wall.[2, 3] We begin by dividing the micelle population into three regions I→III where region I refers to small aggregates such as dimers, trimers, *etc.*, region III refers to micelles proper and region II refers to rare intermediate aggregates whose concentrations are sufficiently small that their contribution to the overall stoichiometry of the system is negligible.

For region I we write

$$\theta_r = \sum_i N_i^r \theta_i \tag{22}$$

whereas for region III

$$\theta_r = \sum_i N_i^r \theta_i + \varepsilon. \tag{23}$$

For region II we write

$$\theta_r = \sum_i N_i^r \theta_i + \varepsilon_r \tag{24}$$

where $\varepsilon_r = 0$ at the boundary between regions I and II and $\varepsilon_r = \varepsilon$ at the boundary between regions II and III. Together with eqn (1) eqn (24) gives for reactions in region II

$$J_{ri} = l_{ri}(\varepsilon_r - \varepsilon_s). \tag{25}$$

There is a strong analogy between this equation and Ohm's law as applied to a resistance between two points r and s. J_{ri} corresponds to the current, ε_r and ε_s correspond to the electrical potentials at r and s and $(l_{ri})^{-1}$ corresponds to the resistance. Moreover,

when the steady-state approximation applies to region II, so that it is legitimate to write for species in this region

$$dC_r/dt = 0 \tag{26}$$

then the rate of formation of micelles of type r from smaller micelles is equal to the rate at which they react to give bigger micelles. This gives us a series of equations between the J_{ri} analogous to Kirchoff's laws. It is clear therefore that for multicomponent micelles the behaviour of region II during the slow process is analogous to that of a network of resistances connecting a region of potential ε with a region of zero potential. Hence we may write

$$J = dC/dt = -l\varepsilon \tag{27}$$

where $-J$ corresponds to the total current through the network, ε corresponds to the total potential drop across it and l is the overall resistance of the network calculable by standard methods from eqn (25) and (26). The negative sign in eqn (27) arises because a positive ε favours a decrease in C_m. Each micellar species in the network corresponds to a junction of resistances, and if there are c micellar constituents then each species connects to c larger species and n smaller species where n, its number of constituents, is $\leqslant c$.

The case of single component micelles is very straightforward. We have simply

$$J_r = -l_r(\varepsilon_r - \varepsilon_{r-1}) \tag{28}$$

but by virtue of eqn (28) $J_r = J$ is the same for all r in region II. Hence

$$J/l_r = -(\varepsilon_r - \varepsilon_{r-1}) \tag{29}$$

and we obtain on summing over all r in the region concerned

$$J = -[\sum_r (l_r)^{-1}]^{-1}\varepsilon = \left(\frac{dC_m}{dt}\right). \tag{30}$$

This expression gives for the slow relaxation time.

$$\frac{1}{\tau_2} = (\sum l_r^{-1})^{-1}\left(\frac{\partial \varepsilon}{\partial C_m}\right)_{C_i} \tag{31}$$

which is a general form of the result obtained previously for ionic surfactants in the presence of supporting electrolyte. It is easily shown that eqn (31) reduces to the expression derived by Aniansson and Wall for the particular case they consider, namely ideal systems which conform to mass-action kinetics with rate coefficients that depend only on T and p. However, the above derivations do not rely on any particular form for either the kinetic equations or the equations describing the dependence of chemical potentials on solution composition. In particular, no assumptions concerning ideality are required.

Consider now eqn (30). This equation is of the linear phenomenological form with overall driving force ε and a phenomenological coefficient which is a combination of the various l_r. The overall coefficient $(\sum_r l_r^{-1})$ is in effect the net resistance of a set of resistances in series. However, it also turns out that $\boldsymbol{RT}(\sum_r l_r^{-1})^{-1}$ may be regarded as the equilibrium rate of micelle formation or breakdown. This latter statement may be justified as follows. Consider an overall reaction which proceeds through a series of intermediates *via* a single route with each step having a stoichiometry number of unity. When the steady-state approximation applies the overall forward and backward rates for such a reaction may be expressed in terms of the rates of the individual steps by writing[23-25]

$$r_f = \prod_{i=1}^{n} \frac{r_i^f}{D} \tag{32a}$$

$$r_b = \prod_{i=1}^{n} \frac{r_i^b}{D} \tag{32b}$$

where i refers to the ith step, n is the total number of steps and

$$D = \prod_{i=2}^{n} r_i^{\mathrm{f}} + r_1^{\mathrm{b}} \prod_{i=3}^{n} r_i^{\mathrm{f}} \ldots \prod_{i=1}^{n-1} r_i^{\mathrm{b}}. \tag{33}$$

At equilibrium when $r_i^{\mathrm{f}} = r_i^{\mathrm{b}}$ for all i we find that r_{f} and r_{b} become

$$r_{\mathrm{f}} = \frac{1}{\sum_i (r_i^{\mathrm{f}})^{-1}} = \frac{1}{\sum_i (r_i^{\mathrm{b}})^{-1}} = r_{\mathrm{b}}. \tag{34}$$

Clearly, since $\boldsymbol{RT}l_r$ is the equilibrium forward or backward rate of the reaction $(r-1)$-mer + monomer $\rightleftharpoons$ r-mer then $\boldsymbol{RT}[\sum (l_r^{-1})]^{-1}$ may be taken as the overall equilibrium rate of micelle formation or breakdown as suggested above.

In some cases it may be legitimate to apply the expressions derived above for single-component micelles to mixed micelles. Consider for example a mixture of two surfactants i and j in which i has a much greater c.m.c. than j. The equilibrium micelle–monomer exchange rate of i can be expected to be correspondingly greater than that of j in both single-component and mixed micelles. Moreover, this should apply for all aggregation numbers and micelle compositions, including the rare intermediate region. Consequently, for a given flux between two adjacent species in this region the driving force will be much less when the two species differ by a molecule of i than when they differ by a molecule of j. Under these circumstances micelles with the same number of j molecules all have approximately the same value of ε irrespective of the amount of i they contain. In other words the change in ε between regions III and I is associated almost entirely with the loss of j molecules. In effect this is equivalent to treating i as a component of a mixed solvent which is assumed either not to participate in the processes under study or to be in complete equilibrium throughout the system during the timescales of interest.

To deal with the situation where C_m can change through the association or disproportionation of aggregates we may proceed as follows. Consider the reaction

$$r\text{-mer} + s\text{-mer} \rightleftharpoons (r+s)\text{-mer}$$

where r and s refer to micellar species in region III. Let the net rate of the reaction be J_{rs}. At chemical equilibrium we have, according to linear phenomenological theory,

$$J_{rs} = l_{rs}\,\varepsilon \tag{35}$$

where $\boldsymbol{RT}l_{rs}$ is the equilibrium rate of the above reaction and ε is the driving force. In the case that r and s are species in region I and $(r+s)$ is a species in region III we have

$$J_{rs} = -l_{rs}\,\varepsilon \tag{36}$$

However, whereas the reaction described by eqn (35) corresponds to the loss of a micelle by region III the reaction described by eqn (36) corresponds to the gain of a micelle by region III. Hence the overall rate of change in C_{m} due to aggregate association and breakdown may be written as

$$\mathrm{d}C_{\mathrm{m}}/\mathrm{d}t = -\sum_{rs} l_{rs}\,\varepsilon \tag{37}$$

where the rs pair may be both in region I or both in region III. Reactions for which r is in region I and s in region III have zero driving force under quasi-equilibrium conditions, and in any event do not contribute to changes in C_{m}, the concentration of aggregates in region III. Reactions involving micelles in region II are considered as sufficiently rare to ignore.

If we now combine eqn (27) and (37) we obtain for the overall rate of change in C_{m} due to both mechanisms acting together

$$\mathrm{d}C_{\mathrm{m}}/\mathrm{d}t = -(l + \sum_{rs} l_{rs})\,\varepsilon. \tag{38}$$

The corresponding relaxation time is given by

$$\frac{1}{\tau_2} = (l + \sum_{rs} l_{rs}) \left(\frac{\partial \varepsilon}{\partial C_m} \right)_{C_i} . \tag{39}$$

Eqn (38) and (39) apply to single- and multi-component micelles alike. For the former eqn (39) is a general and properly justified form of the expression presented previously by Kahlweit and coworkers.[21, 22] For reasons which are not altogether clear Kahlweit and coworkers do not consider reactions between micelles in region III. However, there seems to be no obvious reason why they should not be included.

Eqn (38) suggests strongly that the driving force for changes in C_m at chemical equilibrium is the subdivision potential ε irrespective of the mechanism whereby the changes in C_m take place. Moreover, for single-component micelles at least the coefficient of ε is the equilibrium rate of micelle formation and breakdown. However, it is debatable whether or not talking about reactions such as r-mer + s-mer $\rightleftharpoons$ ($r+s$)-mer is strictly sensible in the present context. Since micelles are in dynamic equilibrium with monomer and the exchange processes concerned are rapid the contents of a given micelle may change appreciably during a reactive encounter with another micelle, especially if the timescale of such encounters is as slow as the values of τ_2 suggest.

Coupling between Fast and Slow Processes

In the discussion so far we have assumed, in common with Aniansson and coworkers and others, that C_m is strictly constant during micelle–monomer exchange and that micelles and monomers are strictly in chemical equilibrium throughout the slow process during which subdivision equilibrium is attained. Clearly these results are not strictly valid and the equations which result from making them are approximate. On intuitive grounds one expects these approximations to be valid when the relaxation times concerned are widely separated. Under these circumstances any changes in C_m which take place during the fast process should be much smaller than those which occur during the slow process. Also the deviation from chemical equilibrium during the slow process should be much smaller than that typically associated with the fast process. These intuitive expectations are supported by some recent model calculations of Kegeles.[26] However, there may be situations in which the above approximations are no longer valid. Two possible examples are (*a*) solutions of shorter chain-length surfactants in which average aggregation numbers are small and species in region II are not that rare and (*b*) solutions of surfactants in apolar media in which small reversed micelles can form. In this section we derive expressions, albeit approximate, which do allow for coupling between the two processes. For simplicity we consider only single-component micelles and as before take as our starting point

$$J_r = -l_r(\theta_r - \theta_{r-1} - \theta_1) \tag{40}$$

where J_r is the net rate of the process

$$(r-1)\text{-mer} + \text{monomer} \rightarrow r\text{-mer}.$$

For the micelles proper in region III we suppose as above that

$$(\theta_r - \theta_{r-1}) = \theta_m \tag{41}$$

is the same for all r and that

$$\theta_r = r\theta_m + \varepsilon. \tag{42}$$

For region I we suppose that premicellar aggregates such as dimers and trimers are in equilibrium with monomer so that

$$\theta_r = r\theta_1. \tag{43}$$

For region II we suppose that

$$(\theta_r - \theta_{r-1} - \theta_1) = A_r. \tag{44}$$

Let this region consist of $(n+1)$ species and let t denote that species with the highest aggregation number. We suppose as before that the steady state approximation holds in this region so that

$$\frac{dC_m}{dt} = J = J_r = l_r A_r \tag{45}$$

where C_m denotes the concentration of aggregates in region III. Summing over the n steps within region II we find that

$$-J(\sum_r l_r^{-1}) = \sum_r A_r = (\theta_t - \theta_{t-n} - \theta_1) \tag{46}$$

since $\theta_t = t\theta_m + \varepsilon$ and $\theta_{t-n} = (t-n)\,\theta_1$ eqn (46) gives

$$\frac{dC_m}{dt} = J = -(\sum l_r^{-1})^{-1}\,\varepsilon - t(\sum_r l_r)^{-1}\,(\theta_m - \theta_1). \tag{47}$$

If, in this context, we regard m_1 as the total concentration in region I then we have for the rate of change of monomer concentration

$$\frac{dm_1}{dt} = -\sum_{III} J_r - nJ - (t-n)J \tag{48}$$

where the summation includes all the species in region III. The second term on the right-hand side of eqn (48) accounts for the n steps in region II and the final term allows for the fact that when a member of species $(t-n)$ breaks down this contributes $(t-n)$ monomers to region I. Substituting for J_r as given by eqn (40) and (41) and for J as given by eqn (47) we obtain

$$\frac{dm_1}{dt} = \left(\sum_{III} l_r + t^2(\sum_{II} l_r^{-1})^{-1}\right)(\theta_m - \theta_1) + t\,(\sum_r l_r^{-1})^{-1}\,\varepsilon. \tag{49}$$

Eqn (47) and (49) constitute two coupled linear phenomenological equations expressing dm_1/dt and dC_m/dt in terms of the two driving forces $(\theta_m - \theta_1)$ and ε. It is noteworthy that the cross-phenomenological coefficients are equal and thus satisfy the Onsager reciprocal relationship.[5]

Clearly the two equations are approximate and it is not immediately obvious whether or not the approximations involved could lead to discrepancies between theory and experiment which may outweigh the effects that arise from the coupling. The equations lead to two relaxation times. To obtain expressions for these we write $(\theta_m - \theta_1)$ and ε in terms of the perturbations in m_1 and C_m, *i.e.*

$$(\theta_m - \theta_1) = \left(\frac{\partial\theta_m - \theta_1}{\partial m_1}\right)_{C_m,\,C_1} \Delta m_1 + \left(\frac{\partial\theta_m - \theta_1}{\partial C_m}\right)_{m_1,\,C_1} \Delta C_m \tag{50a}$$

$$\varepsilon = \left(\frac{\partial\varepsilon}{\partial m_1}\right)_{C_m,\,C_1} \Delta m_1 + \left(\frac{\partial\varepsilon}{\partial C_m}\right)_{m_1,\,C_1} \Delta C_m \tag{50b}$$

substituting these expressions into eqn (48) and (49) we obtain

$$\left.\begin{aligned} \frac{d\Delta m_1}{dt} &= -a_{11}\,\Delta m_1 - a_{1m}\,\Delta C_m \qquad (51a)\\ \frac{d\Delta C_m}{dt} &= -a_{m1}\,\Delta m_1 - a_{mm}\,\Delta C_m \qquad (51b) \end{aligned}\right\}$$

where

$$a_{11} = -[\sum_{\text{III}} l_r + t^2(\sum_{\text{II}} l_r^{-1})^{-1}]\left(\frac{\partial\theta_m - \theta_1}{\partial m_1}\right)_{C_m, C_1} - t(\sum_{\text{II}} l_r^{-1})^{-1}\left(\frac{\partial\varepsilon}{\partial m_1}\right)_{C_m, C_1} \quad (52a)$$

$$a_{1m} = -[\sum_{\text{III}} l_r + t^2(\sum_{\text{II}} l_r^{-1})^{-1}]\left(\frac{\partial\theta_m - \theta_1}{\partial C_m}\right)_{m_1, C_1} - t(\sum_{\text{II}} l_r^{-1})^{-1}\left(\frac{\partial\varepsilon}{\partial C_m}\right)_{m_1, C_1} \quad (52b)$$

$$a_{m1} = t(\sum_{\text{II}} l_r^{-1})^{-1}\left(\frac{\partial\theta_m - \theta_1}{\partial m_1}\right)_{C_m, C_1} + (\sum_{\text{II}} l_r^{-1})^{-1}\left(\frac{\partial\varepsilon}{\partial m_1}\right)_{C_m, C_1} \quad (52c)$$

$$a_{mm} = t(\sum_{\text{II}} l_r^{-1})^{-1}\left(\frac{\partial\theta_m - \theta_1}{\partial C_m}\right)_{m_1, C_1} + (\sum_{\text{II}} l_r^{-1})^{-1}\left(\frac{\partial\varepsilon}{\partial C_m}\right)_{m_1, C_1}. \quad (52d)$$

The two relaxation times are given by

$$\frac{1}{\tau_1} = \frac{a_{11} + a_{mm}}{2} + \left(\frac{(a_{11} - a_{mm})^2 + a_{1m}\, a_{m1}}{4}\right)^{\frac{1}{2}} \quad (53a)$$

$$\frac{1}{\tau_2} = \frac{a_{11} + a_{mm}}{2} - \left(\frac{(a_{11} + a_{mm})^2 + a_{1m}\, a_{m1}}{4}\right)^{\frac{1}{2}}. \quad (53b)$$

For solutions of non-interacting aggregates calculation of the thermodynamic terms in the above expressions is straightforward; also, when $a_{11} > a_{mm}$ eqn (53a) and (53b) become equivalent to the expressions which result when coupling between the fast and slow processes is ignored.

Non-linear Treatment of Fast and Slow Processes

For many systems of practical interest the perturbations from equilibrium are too large for the linear theory described above to apply. Hence it is important to establish to what extent one can develop fairly general arguments in the non-linear regime. A suitable starting point for such procedures is the expression

$$J_{ri}^{f}/J_{ri}^{b} = \exp\frac{(\theta_r + \theta_i - \theta_s)}{RT} \quad (54)$$

where J_{ri}^{f} and J_{ri}^{b} are the forward and backward rates of the reaction

$$r + i \rightleftharpoons s.$$

Eqn (54) is equally applicable to ideal and non-ideal solutions.[11] It leads to the result

$$J_{ri} = (J_{ri}^{f} - J_{ri}^{b}) = J_{ri}^{b}\exp\left(\frac{(\theta_r + \theta_i - \theta_s)}{RT}\right) - 1 = J_{ri}^{f}\left(1 - \exp\frac{(\theta_s - \theta_r - \theta_i)}{RT}\right). \quad (55)$$

Eqn (55) is a general non-linear form of eqn (1) and becomes identical to this expression when it is legitimate to linearise the exponential.

Fast Process

As before

$$\frac{dm_i}{dt} = -\sum_r J_{ri} \quad (56)$$

which together with eqn (55) gives

$$\frac{dm_i}{dt} = -\sum_r J_{ri}^{f}\left(1 - \exp\frac{(\theta_s - \theta_r - \theta_i)}{RT}\right). \quad (57)$$

When $(\theta_s-\theta_r)=\theta_i^{\mathrm{m}}$ eqn (57) becomes

$$\frac{\mathrm{d}m_i}{\mathrm{d}t}=-(\Sigma J_{ri}^{\mathrm{f}})\left(1-\exp\frac{(\theta_i^{\mathrm{m}}-\theta_i)}{\boldsymbol{RT}}\right). \tag{58}$$

For single-component micelles in ideal systems it is probably quite a good approximation to put

$$J_{ri}^{\mathrm{f}}=k_f C_{r-1} m_1 \tag{59}$$

where k_f is the same for all micellar species. This gives

$$\frac{\mathrm{d}m_1}{\mathrm{d}t}=k_f C_{\mathrm{m}} m_1\left(\exp\frac{\theta_{\mathrm{m}}-\theta_1}{\boldsymbol{RT}}-1\right). \tag{60}$$

However,

$$(\theta_{\mathrm{m}}-\theta_1)=(\theta_{\mathrm{m}}-\theta^*)-(\theta_1-\theta^*) \tag{61}$$

and in this case

$$(\theta_1-\theta^*)=\boldsymbol{RT}\ln m_1/m_1^* \tag{62}$$

where m_1^* is the quasi-equilibrium value of m_1, which may be regarded as a known quantity. Together with eqn (60) eqn (62) gives

$$\frac{\mathrm{d}m_1}{\mathrm{d}t}=k_f C_{\mathrm{m}}\left(m_1^*\exp\frac{\theta_{\mathrm{m}}-\theta^*}{\boldsymbol{RT}}-m_1\right) \tag{63}$$

now

$$\theta_{\mathrm{m}}-\theta^*=\int_{C_1^{\mathrm{m}*}}^{C_1^{\mathrm{m}}}\left(\frac{\partial\theta_{\mathrm{m}}}{\partial C_1^{\mathrm{m}}}\right)_{C_{\mathrm{m}}}\mathrm{d}C_1^{\mathrm{m}} \tag{64}$$

and

$$\left(\frac{\partial\theta_{\mathrm{m}}}{\partial C_1^{\mathrm{m}}}\right)_{C_{\mathrm{m}}}=\frac{\boldsymbol{RT}}{\sigma^2 C_{\mathrm{m}}}. \tag{65}$$

Thus if σ^2 is known as a function of C_1^{m} at constant C_{m} the integral on the right-hand side of eqn (64) can be expressed in terms of the departure from equilibrium and we obtain a non-linear first-order differential equation relating $\mathrm{d}m_1/\mathrm{d}t$ to some function of m_1. However, it should be stressed that this equation is approximate because of the assumptions that $(\theta_r-\theta_{r-1})$ and k_r^{f} are the same for all micellar species.

Slow Process

Stepwise Association Mechanism

As above we divide the aggregate population into three regions. In region II during the slow process we have

$$J_{ri}^{\mathrm{f}}/J_{ri}^{\mathrm{b}}=\exp\left[(\varepsilon_r-\varepsilon_s)/\boldsymbol{RT}\right] \tag{66}$$

where

$$\left.\begin{aligned}\varepsilon_r&=\theta_r-\Sigma N_i^r\theta_i\\ \varepsilon_s&=\theta_s-\Sigma N_i^s\theta_i\end{aligned}\right\} \tag{67a,b}$$

so that

$$\varepsilon_r-\varepsilon_s=(\theta_r+\theta_i-\theta_s).$$

It follows that

$$J_{ri}=J_{ri}^{\mathrm{b}}\left(\exp-\frac{(\varepsilon_s-\varepsilon_r)}{\boldsymbol{RT}}-1\right)=J_{ri}^{\mathrm{b}}\exp-\frac{\varepsilon_s}{\boldsymbol{RT}}\left[\exp\left(\frac{\varepsilon_r}{\boldsymbol{RT}}\right)-\exp\left(\frac{\varepsilon_s}{\boldsymbol{RT}}\right)\right]. \tag{68}$$

When the steady-state approximation is valid eqn (18) is in some ways analogous to eqn (25). Again J_{ri} is analogous to a current and $\exp\varepsilon_r/\boldsymbol{kT}$ is analogous to a potential at some junction r. The term $J_{ri}^{\mathrm{b}}\exp-\varepsilon_s/\boldsymbol{kT}$ plays the same role as l_{ri}. For solutions in which all

monomer and aggregate species behave ideally $J_{ri}^{b} = k_{si}^{b} C_s$ where k_{si}^{b} is the rate constant for the loss of an i molecule by species s and C_s is the concentration of species s given by

$$C_s \exp -(\varepsilon_s/\boldsymbol{RT}) = C_s^{e} \tag{69}$$

where C_s^{e} is the concentration of species s in the solution at complete equilibrium in which the monomer concentrations are the same as those in the reacting system of interest. Hence $J_{ri}^{b} \exp(-\varepsilon_s/\boldsymbol{kT})$ has the significance of an equilibrium rate. Clearly the non-linear case can be regarded as analogous to a network of resistances in much the same way as the linear case.

For micelles of a single component eqn (68) becomes

$$\frac{dC_m}{dt} = J = J_r = J_r^{b} \exp -(\varepsilon_r/\boldsymbol{RT}) [\exp(\varepsilon_{r-1}/\boldsymbol{RT}) - \exp(\varepsilon_r/\boldsymbol{RT})] \tag{70}$$

which is the non-linear analogue of eqn (28). Dividing both sides by $J_r^{b} \exp -(\varepsilon_r/\boldsymbol{RT})$ and summing overall r in region II we obtain

$$J \sum_r (J_r^{b})^{-1} \exp(\varepsilon_r/\boldsymbol{RT}) = [1 - \exp(\varepsilon/\boldsymbol{RT})] \tag{71}$$

so that

$$J = \frac{dC_m}{dt} - [\sum_r (J_r^{b})^{-1} \exp(\varepsilon_r/\boldsymbol{RT})]^{-1} [1 - \exp(\varepsilon/\boldsymbol{RT})]. \tag{72}$$

By use of eqn (31), (33) and (66) it is straightforward to show that the term $[\sum_r (J_r^{b})^{-1} \exp(\varepsilon_r/\boldsymbol{RT})]^{-1} \exp(\varepsilon/\boldsymbol{RT})$ is the rate of micelle breakdown. If we denote this quantity by J_m^{b} and the corresponding quantity for the rate of micelle formation by J_m^{f} then it also follows that

$$\frac{J_m^{b}}{J_m^{f}} = \exp(\varepsilon/\boldsymbol{RT}). \tag{73}$$

Indeed these results are particular cases of a more general relationship described elsewhere.[11, 25]

For ideal systems eqn (72) gives

$$J = \frac{dC_m}{dt} = [\sum_r (k_r^{b} C_r^{e})^{-1}]^{-1} [1 - \exp(\varepsilon/\boldsymbol{RT})] \tag{74}$$

but in this case

$$\exp(\varepsilon/\boldsymbol{RT}) = C_m/C_m^{e} \tag{75}$$

where C_m^{e} is the concentration of micelles in a system at complete equilibrium with the same monomer concentration as in the reacting system. Hence

$$J = \frac{dC_m}{dt} = [\sum_r (k_r^{b} C_r^{e})^{-1}]^{-1} [1 - (C_m/C_m^{e})]. \tag{76}$$

However, since $C_m = (C_1 - m_1)/N$ and $\bar{N} = \bar{N}(m_1)$ under the conditions of interest it follows that the right hand side of eqn (76) is a function of m_1 only. The left-hand side may now be written as $(\partial C_m/\partial m_1)_{C_1} \, dm_1/dt$ and it is straightforward to show that

$$\left(\frac{\partial C_m}{\partial m_1}\right)_{C_1} = -\left(\frac{\sigma^2 C_1 + (\bar{N} - \sigma^2) m_1}{\bar{N}^2 m_1}\right). \tag{77}$$

Hence we obtain finally

$$\frac{dm_1}{dt} = \frac{\bar{N} m_1 (C_1 - C_1^{e})}{[\sigma^2 C_1 + (\bar{N} - \sigma^2) m_1] \, C_m^{e} \sum_r [k_r^{b} C_r^{e}]^{-1}} \tag{78}$$

where C_e is the value of C in the system at complete equilibrium with the m_1 concerned.

Eqn (78) expresses dm_1/dt as a function of m_1 only and can be integrated to obtain m_1 and hence C_m as functions of t when explicit expressions in terms of m_1 for the various quantities on the right-hand side are available.

When micelle association and disproportionation can occur we have instead of eqn (36)

$$J^f_{rs}/J^b_{rs} = \exp(\varepsilon/\boldsymbol{RT}) \tag{79}$$

which gives

$$J_{rs} = J^b_{rs}[\exp(\varepsilon/\boldsymbol{RT}) - 1]. \tag{80}$$

Eqn (79) and (80) apply to reactions between micelles in region III.

For association of micelles in region I to give micelles in region III we have

$$J^f_{rs}/J^b_{rs} = \exp(-\varepsilon/\boldsymbol{RT}). \tag{81}$$

The contribution of micellar association to dC_m/dt is thus given by

$$-\sum_{rs}^{III} J_{rs} + \sum_{rs}^{I} J_{rs} = \sum_{rs} J^b_{rs}[\exp(\varepsilon/\boldsymbol{RT}) - 1]. \tag{82}$$

For ideal systems

$$J^b_{rs} = k^b_{rs} C_{(r+s)} = k^b_{rs} C^e_{(r+s)} \exp(\varepsilon/\boldsymbol{RT}). \tag{83}$$

Hence

$$\frac{dC_m}{dt} = -\sum_{rs} k^b_{rs} C^e_{(r+s)} \exp(\varepsilon/\boldsymbol{RT})[\exp(\varepsilon/\boldsymbol{RT}) - 1]$$

$$= -\sum_{rs} k^f_{rs} C^e_r C^e_s \exp(\varepsilon/\boldsymbol{RT})[\exp(\varepsilon/\boldsymbol{RT}) - 1] \tag{84}$$

where C^e_r C^e_s and $C^e_{(r+s)}$ are the concentrations of the species concerned in a system at total equilibrium where the monomer concentrations are the same as in the reacting system of interest.

Combining eqn (72) and (82) we find that when both mechanisms for changes in C_m operate

$$\frac{dC_m}{dt} = -J^b_m[1 - \exp-(\varepsilon/\boldsymbol{RT})] \tag{85}$$

where J^b_m is the rate of micelle disappearance. In this context it must be remembered that the reaction

$$r_{mer} + s_{mer} \rightleftharpoons (r+s)\text{-mer}$$

constitutes micelle disappearance when proceeding from left to right.

Concluding Remarks

The above examples illustrate well the power and generality of phenomenological methods in describing the kinetics of micellisation. A particular advantage of the treatment is that the thermodynamic and kinetic terms can be handled separately. Hence the treatment is not restricted to any particular model for the equilibrium thermodynamic behaviour. In some respects the approach is more convenient than traditional methods based on mass-action kinetics. This arises primarily from the fact that the rates of the individual steps may be written as a single term rather than as the difference between two terms. The approach is also more general than previous treatments, and indeed all the significant results obtained previously follow straightforwardly when appropriate expressions for the thermodynamic and kinetic terms are inserted into the general equations.

A number of important advances have been made. These include (i) the treatment of mixed micelles which leads to expressions that enable equilibrium thermodynamic

data to be used effectively in the interpretation of relaxation experiments, (ii) the demonstration that the subdivision potential is the driving force for changes in the micelle number during the slow process both for the stepwise association mechanism and the mechanism of aggregate association and disproportionation, (iii) the recognition that the phenomenological coefficient for the slow process may be regarded as the equilibrium rate of micelle formation or breakdown divided by RT and (iv) the derivation of equations which allow for coupling between the fast and slow processes.

In the last section perturbations from equilibrium which are too large to handle using the linear formalism were considered. As might have been expected the general formal expressions obtained are difficult to handle without making considerable simplifications. However, they do provide a basis for further work.

Appendix

The Approximation in eqn (8)

The approximation in eqn (8) consists of replacing the micelle population in the real relaxing system with that population which minimises g for the same values of the m_k, C_k^{m} and C_{m}. For single-component micelles we may argue quantitatively as follows. Eqn (6) takes the form

$$\frac{\mathrm{d}m_1}{\mathrm{d}t} = \sum_r l_r(\theta_r - \theta_{r-1} - \theta_1) = \sum_r l_r[(\theta_r - \theta_r^*) - (\theta_{r-1} - \theta_{r-1}^*) - (\theta_1 - \theta_1^*)] \tag{A1}$$

where l_r is the phenomenological coefficient for the reaction

$$(r-1)\text{-mer} + \text{monomer} \rightleftharpoons r\text{-mer}.$$

Eqn (A1) may be written in the form

$$\frac{\mathrm{d}m_1}{\mathrm{d}t} = \sum_r X_r \Delta C_r + X_1 \Delta m_1 \tag{A2}$$

where

$$X_r = \sum_s l_s \left[\left(\frac{\partial \theta_s}{\partial C_r} \right)_{C_t,\, m_1} - \left(\frac{\partial \theta_{s-1}}{\partial C_r} \right)_{C_t,\, m_1} - \left(\frac{\partial \theta_1}{\partial C_r} \right)_{C_t,\, m_1} \right] \tag{A3a}$$

$$X_1 = \sum_s l_s \left[\left(\frac{\partial \theta_s}{\partial m_1} \right)_{C_r} - \left(\frac{\partial \theta_{s-1}}{\partial m_1} \right)_{C_r} - \left(\frac{\partial \theta_1}{\partial m_1} \right)_{C_r} \right]. \tag{A3b}$$

All derivatives are taken at constant T, p and C_{m} and all refer to that state of the system at the end of the fast process. C_t denotes that all C_s other than the differentiating variable are held constant.

Let us regard X_r as a continuous function of r, the number of surfactant molecules per micelle. We may replace the summation in eqn (A2) by an integral so that

$$\sum_r X_r \Delta C_r = \int_{r_{\min}}^{\infty} X(r)\,(C_r - C_r^*)\,\mathrm{d}r \tag{A4}$$

where the lower limit of integration corresponds to the smallest aggregation number we may reasonably call micellar. We may now expand $X(r)$ as a Taylor series about some value of r, r^+ which we may if we wish take as the most probable value of r. This gives

$$X(r) = X^+ + \left(\frac{\partial X}{\partial r} \right)_{r=r^+} (r - r^+) + \frac{1}{2} \left(\frac{\partial^2 X}{\partial r^2} \right)_{r=r^+} (r - r^+)^2 + \ldots \tag{A5}$$

so that together with eqn (A4) we obtain

$$\sum X_r \Delta C_r = X^+ \int (C_r - C_r^*)\,\mathrm{d}r + \left(\frac{\partial X}{\partial r}\right)_{r=r^+} \int (r-r^+)(C_r - C_r^*)\,\mathrm{d}r + \frac{1}{2}\left(\frac{\partial^2 X}{\partial r^2}\right)_{r=r^+} \int (r-r^+)^2 (C_r - C_r^*)\,\mathrm{d}r + \ldots. \quad \text{(A6)}$$

Since $C_m = \sum_r C_r$ is constant it follows that the first term on the right-hand side of eqn (A6) is zero. Also

$$\int r(C_r - C_r^*)\,\mathrm{d}r = C_m \Delta \bar{N} \quad \text{(A7)}$$

where $\Delta \bar{N}$ is the perturbation in the mean aggregation number and

$$\int r^2 (C_r - C_r^*)\,\mathrm{d}r = C_r \Delta \overline{N^2} \quad \text{(A8)}$$

where $\Delta \overline{N^2}$ is the perturbation in the mean-square aggregation number. Hence we may now write

$$\sum_r X_r \Delta C_r = \left[\left(\frac{\partial X}{\partial r}\right)_{r=r^+} - r^+ \left(\frac{\partial^2 X}{\partial r^2}\right)_{r=r^+}\right] C_m \Delta \bar{N} + \frac{1}{2}\left(\frac{\partial^2 X}{\partial r^2}\right)_{r=r^+} C_m \Delta \overline{N^2} + \text{higher-order terms.} \quad \text{(A9)}$$

Consider now the second term on the right-hand side of eqn (7). For the case of interest this becomes

$$\sum_r l_r[(\theta_r - \theta_r') - (\theta_{r-1} - \theta_{r-1}') - (\theta_1 - \theta_r')]$$

and may be written as

$$\sum_r X_r \Delta C_r'$$

where $\Delta C_r' = (C_r - C_r')$.

Using the same procedures as above we may now write

$$\sum_r X_r \Delta C_r' = X^+ \int (C_r - C_r')\mathrm{d}r + \left(\frac{\partial X}{\partial r}\right)_{r=r^+} \int (r-r^+)(C_r - C_r')\,\mathrm{d}r + \frac{1}{2}\left(\frac{\partial^2 X}{\partial r^2}\right)_{r=r^+} \int (r-r^+)^2 (C_r - C_r')\,\mathrm{d}r. \quad \text{(A10)}$$

However, in this case C_m and $\bar{N}C_m$ are the same for both populations so that

$$\int (C_r - C_r')\,\mathrm{d}r = 0$$

and

$$\int (r-r^+)(C_r - C_r')\,\mathrm{d}r = 0.$$

Hence

$$\sum_r X_r \Delta C_r' = \frac{1}{2}\left(\frac{\partial^2 X}{\partial r^2}\right)_{r=r^+} \int r^2 (C_r - C_r')\,\mathrm{d}r + \text{higher-order terms} = \frac{1}{2}\left(\frac{\partial^2 X}{\partial r^2}\right)_{r=r^+} C_m [\overline{N^2} - (\bar{N})^2] + \text{higher-order terms.} \quad \text{(A11)}$$

It follows therefore that eqn (8) is a good approximation for single-component micelles when $\partial^2 X/\partial r^2$ and higher-order derivatives are all very small. To show the equivalence of this discussion with that of Almgren *et al.*[4] we note that for non-interacting micelles $(\partial\theta_s/\partial C_r) = 0$ unless $r=s$, in which case $(\partial\theta_r/\partial C_r) = \boldsymbol{RT}/C_r$. Thus

$$X_r = l_r\frac{\boldsymbol{RT}}{C_r} - l_{r+1}\frac{\boldsymbol{RT}}{C_r}. \tag{A12}$$

However
$$l_r = \frac{k_r^{\mathrm{b}} C_r}{\boldsymbol{RT}} = \frac{k_r^{\mathrm{f}} C_{r-1} m_1}{\boldsymbol{RT}}$$

where k_r^{f} and k_r^{b}, respectively, are forward and backward rate coefficients for the reaction $(r-1)$-mer + monomer $\rightleftharpoons$ r-mer. Hence

$$X_r = k_r^{\mathrm{b}} - k_{r+1}^{\mathrm{f}} m_1. \tag{A13}$$

Since m_1, the equilibrium value of the monomer concentration, depends on the total concentration, the condition that the second and higher derivatives of X_r with respect to r vanish under all circumstances (*i.e.* for arbitrary perturbations at all concentrations) is that k^{b} and k^{f} are both linear functions of r. This is the result given by Almgren *et al.*[4]

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